Learn Python

Brief History of Python

Invented in the Netherlands, early 90s by Guido van Rossum

Named after Monty Python

Open sourced from the beginning

Considered a scripting language, but is much more

Scalable, object oriented and functional from the beginning

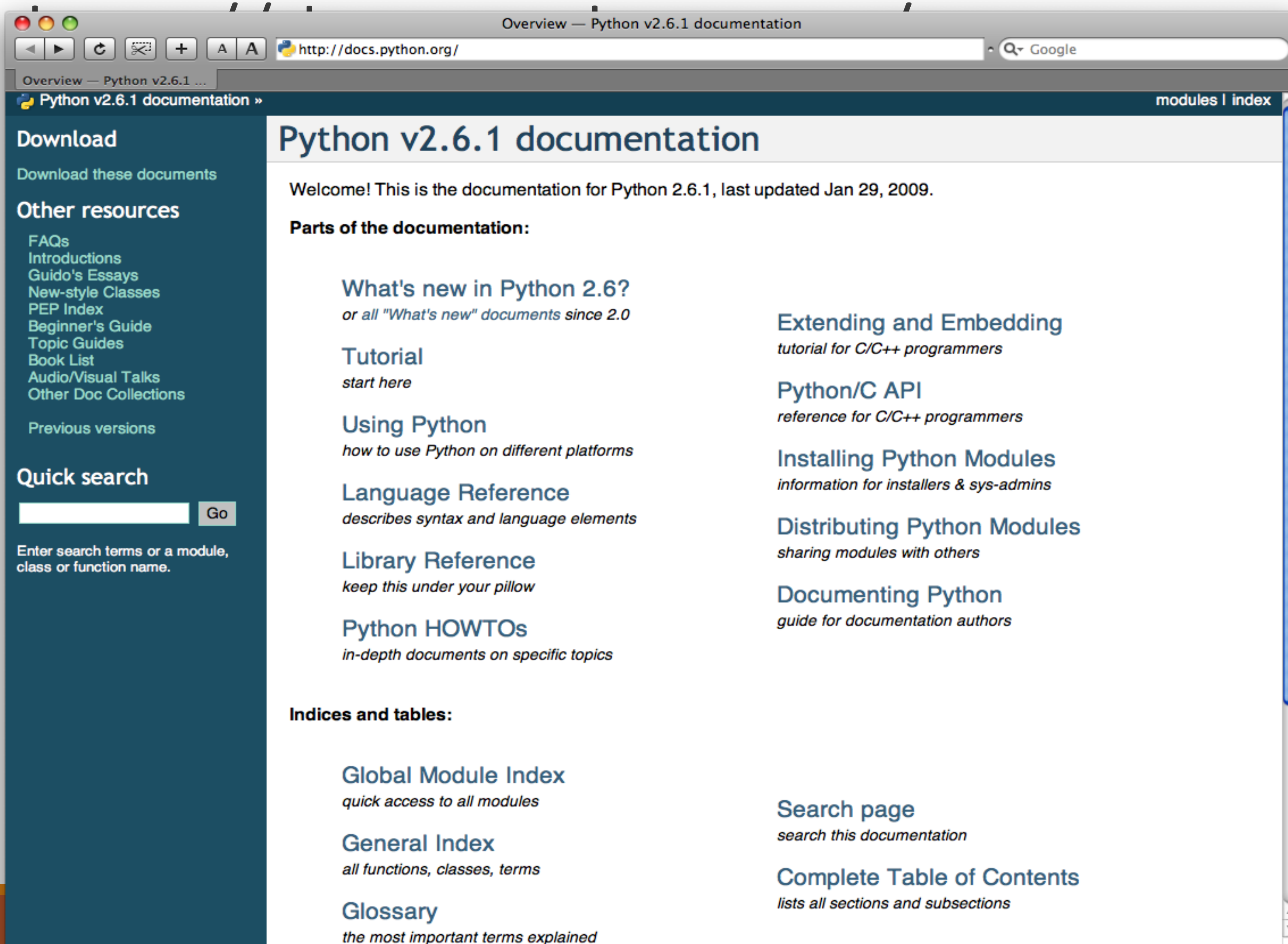
Used by Google from the beginning

Increasingly popular

Python’s Benevolent Dictator For Life

“Python is an experiment in how much freedom programmers need. Too much freedom and nobody can read another's code; too little and expressiveness is endangered.”

- Guido van Rossum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Using Python
how to use Python on different platforms
Language Reference
describes syntax and language elements
Library Reference
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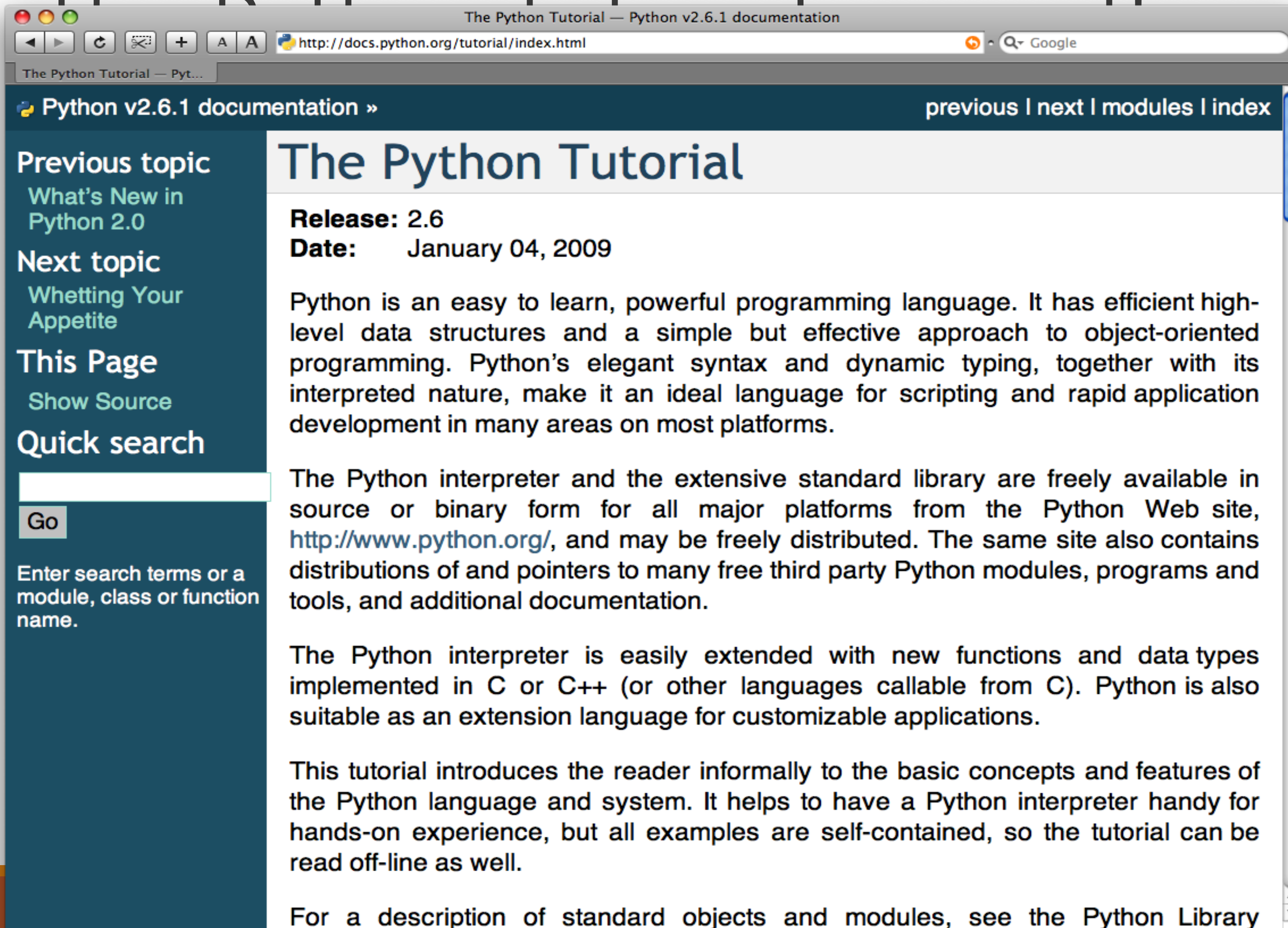

The Python Tutorial — Python v2.6.1 documentation

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The Python Tutorial

Release: 2.6
Date: January 04, 2009

Python is an easy to learn, powerful programming language. It has efficient high-level data structures and a simple but effective approach to object-oriented programming. Python's elegant syntax and dynamic typing, together with its interpreted nature, make it an ideal language for scripting and rapid application development in many areas on most platforms.

The Python interpreter and the extensive standard library are freely available in source or binary form for all major platforms from the Python Web site, http://www.python.org/, and may be freely distributed. The same site also contains distributions of and pointers to many free third party Python modules, programs and tools, and additional documentation.

The Python interpreter is easily extended with new functions and data types implemented in C or C++ (or other languages callable from C). Python is also suitable as an extension language for customizable applications.

This tutorial introduces the reader informally to the basic concepts and features of the Python language and system. It helps to have a Python interpreter handy for hands-on experience, but all examples are self-contained, so the tutorial can be read off-line as well.

For a description of standard objects and modules, see the Python Library

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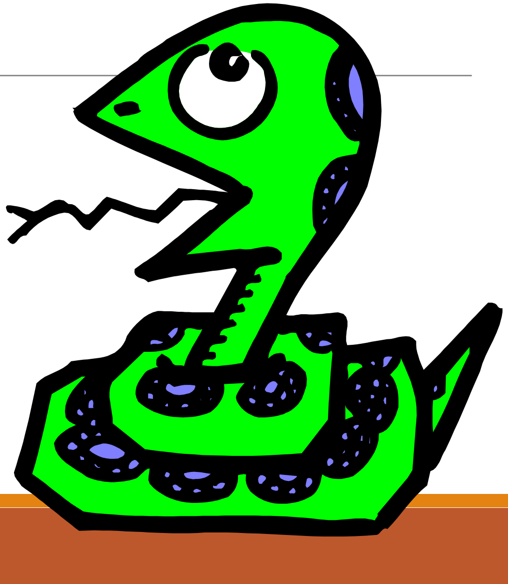
Running Python



The Python Interpreter

Typical Python implementations offer both an interpreter and compiler

Interactive interface to Python with a read-eval-print loop

```
[finin@linux2 ~]$ python
Python 2.4.3 (#1, Jan 14 2008, 18:32:40)
[GCC 4.1.2 20070626 (Red Hat 4.1.2-14)] on linux2
Type "help", "copyright", "credits" or "license" for more information.
>>> def square(x):
...   return x * x
...
>>> map(square, [1, 2, 3, 4])
[1, 4, 9, 16]
>>>
```

Installing

Python is pre-installed on most Unix systems, including Linux and MAC OS X

The pre-installed version may not be the most recent one (3.12)

Download from http://python.org/download/

Python comes with a large library of standard modules

There are several options for an IDE

- IDLE – works well with Windows
- VS CODE
- Eclipse with Pydev (http://pydev.sourceforge.net/)

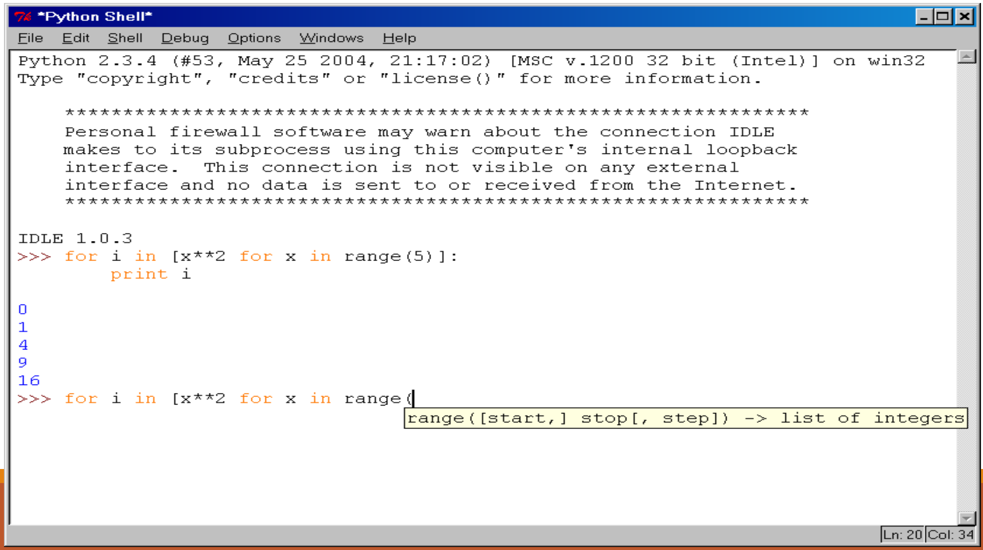
IDLE Development Environment

IDLE is an Integrated DeveLopment Environ-ment for Python, typically used on Windows

Multi-window text editor with syntax highlighting, auto-completion, smart indent and other.

Python shell with syntax highlighting.

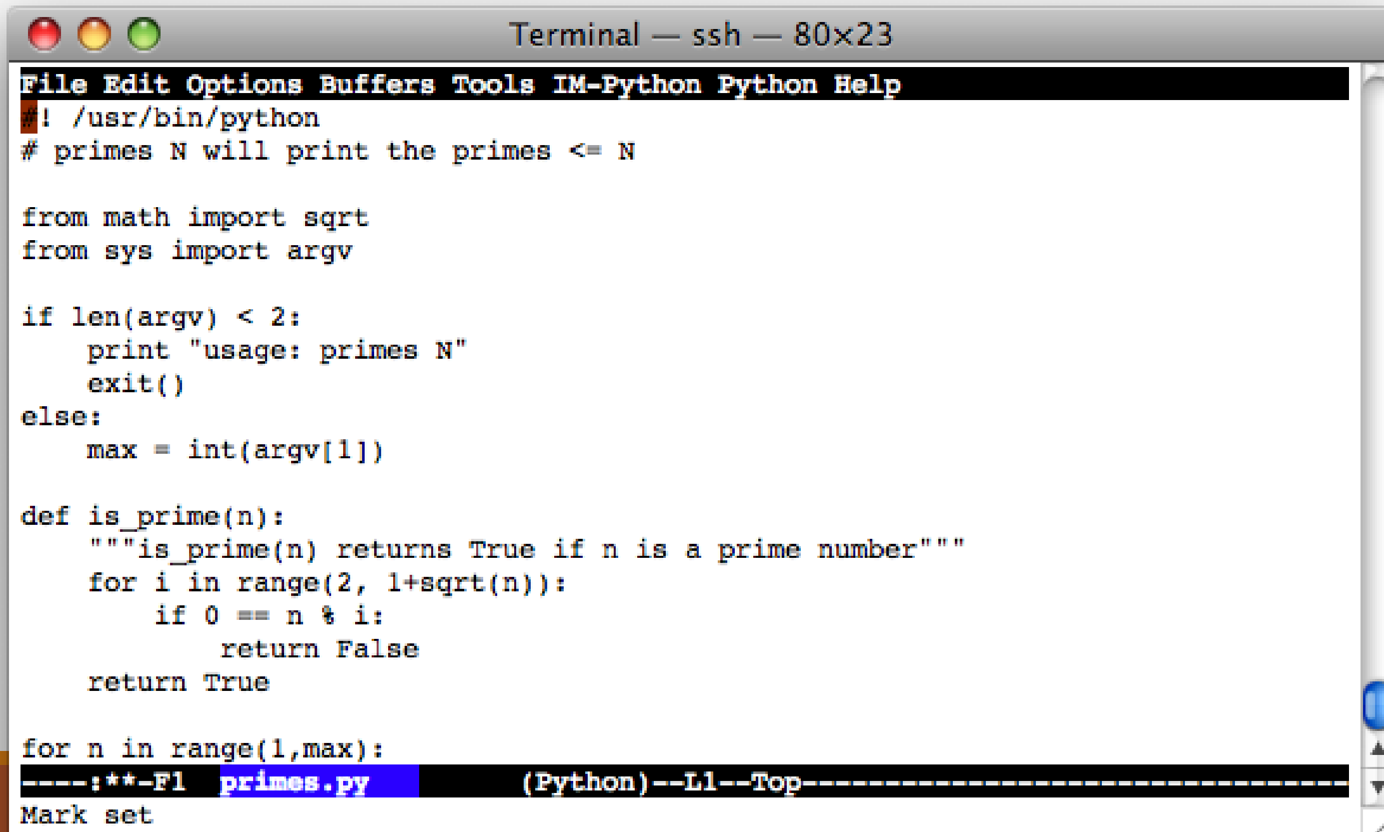
Integrated debugger with stepping, persis-tent breakpoints, and call stack visi-bility



Editing Python in Emacs

Emacs *python-mode* has good support for editing Python, enabled enabled by default for .py files

Features: completion, symbol help, eldoc, and inferior interpreter shell, etc.



Running Interactively on UNIX

On Unix…

```
% python
>>> 3+3
6
```

Python prompts with ‘>>>’.

To exit Python (not Idle):

- In Unix, type CONTROL-D
- In Windows, type CONTROL-Z + <Enter>
- Evaluate exit()

Running Programs on UNIX

Call python program via the python interpreter

```
% python fact.py
```

Make a python file directly executable by

- Adding the appropriate path to your python interpreter as the first line of your file

  ```
  #!/usr/bin/python
  ```

- Making the file executable

  ```
  % chmod a+x fact.py
  ```

- Invoking file from Unix command line

  ```
  % fact.py
  ```

Example ‘script’: fact.py

#! /usr/bin/python

```
def fact(x):
  """Returns the factorial of its argument, assumed to be a posint"""
    if x == 0:
        return 1
    return x * fact(x - 1)

print
print ’N fact(N)’
print "---------"

for n in range(10):
    print n, fact(n)
```

Python Scripts

When you call a python program from the command line the interpreter evaluates each expression in the file

Familiar mechanisms are used to provide command line arguments and/or redirect input and output

Python also has mechanisms to allow a python program to act both as a script and as a module to be imported and used by another python program

Example of a Script

```
#! /usr/bin/python

""" reads text from standard input and outputs any email
  addresses it finds, one to a line.
"""

import re
from sys import stdin

# a regular expression ~ for a valid email address
pat = re.compile(r'[-\w][-.\w]*@[-\w][-\w.]+[a-zA-Z]{2,4}')

for line in stdin.readlines():
    for address in pat.findall(line):
        print address
```

results

```
python> python email0.py <email.txt
bill@msft.com
gates@microsoft.com
steve@apple.com
bill@msft.com
python>
```

Getting a unique, sorted list

```
import re

from sys import stdin

pat = re.compile(r'[-\w][-.\w]*@[-\w][-\w.]+[a-zA-Z]{2,4}')

# found is an initially empty set (a list w/o duplicates)

found = set( )

for line in stdin.readlines():

    for address in pat.findall(line):

        found.add(address)

# sorted() takes a sequence, returns a sorted list of its elements

for address in sorted(found):
```

results

```
python> python email2.py <email.txt
bill@msft.com
gates@microsoft.com
steve@apple.com
python>
```

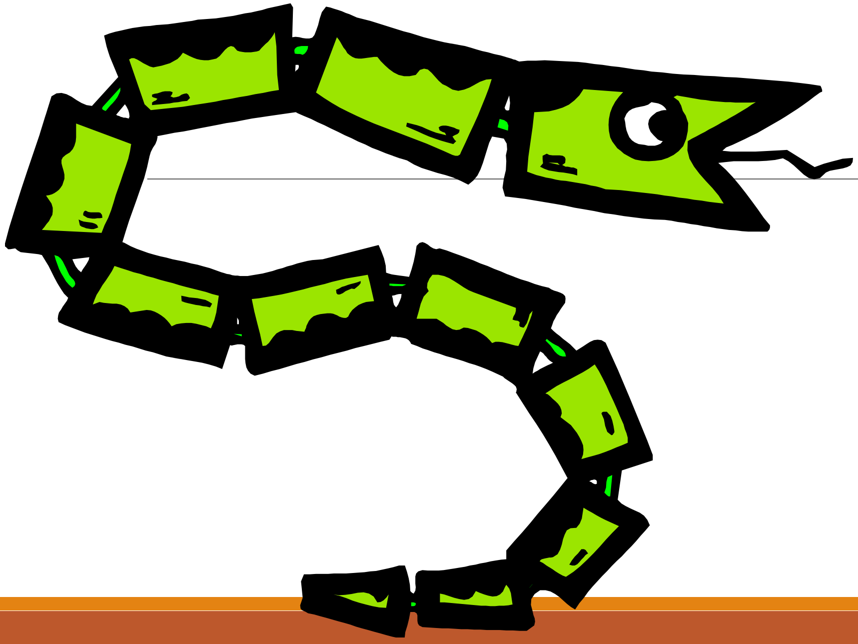
Simple functions: ex.py

```
"""factorial done recursively and iteratively"""

def fact1(n):

    ans = 1

    for i in range(2,n):

        ans = ans * n

    return ans



def fact2(n):

    if n < 1:

        return 1
```

Simple functions: ex.py

```
671> python
Python 2.5.2 …
>>> import ex
>>> ex.fact1(6)
1296
>>> ex.fact2(200)
78865786736479050355236321393218507…000000L
>>> ex.fact1
<function fact1 at 0x902470>
>>> fact1
Traceback (most recent call last):
```

The Basics



A Code Sample (in IDLE)

```
x = 34 - 23             # A comment.
y = “Hello”             # Another one.
z = 3.45
if z == 3.45 or y == “Hello”:
    x = x + 1
    y = y + “ World”    # String concat.
print x
print y
```

Enough to Understand the Code

Indentation matters to code meaning

- Block structure indicated by indentation

First assignment to a variable creates it

- Variable types don't need to be declared.
- Python figures out the variable types on its own.

Assignment is = and comparison is ==

For numbers + - * / % are as expected

- Special use of **+** for string concatenation and **%** for string formatting (as in C's printf)

Logical operators are words (`and, or, not`) *not* symbols

The basic printing command is `print`

Basic Datatypes

Integers (default for numbers)

```
z = 5 / 2  # Answer 2, integer division
```

Floats

```
x = 3.456
```

Strings

- Can use “” or ‘’ to specify with `“abc” == ‘abc’`
- Unmatched can occur within the string: `“matt’s”`
- Use triple double-quotes for multi-line strings or strings than contain both ‘ and “ inside of them: `“““a‘b“c”””`

Whitespace

Whitespace is meaningful in Python: especially indentation and placement of newlines

Use a newline to end a line of code

- Use \ when must go to next line prematurely

No braces { } to mark blocks of code, use *consistent* indentation instead

- First line with *less* indentation is outside of the block
- First line with *more* indentation starts a nested block

Colons start of a new block in many constructs, e.g. function definitions, then clauses

Comments

Start comments with #, rest of line is ignored

Can include a “documentation string” as the first line of a new function or class you define

Development environments, debugger, and other tools use it: it’s good style to include one

```
def fact(n):
  """fact(n) assumes n is a positive
 integer and returns facorial of n."""
 assert(n>0)
 return 1 if n==1 else n*fact(n-1)
```

Assignment

Binding a variable in Python means setting a *name* to hold a *reference* to some *object*

- *Assignment creates references, not copies*

Names in Python do not have an intrinsic type, objects have types

- Python determines the type of the reference automatically based on what data is assigned to it

You create a name the first time it appears on the left side of an assignment expression:

```
x = 3
```

A reference is deleted via garbage collection after any names bound to it have passed out of scope

Python uses *reference semantics* (more later)

Naming Rules

Names are case sensitive and cannot start with a number. They can contain letters, numbers, and underscores.

```
bob   Bob   _bob   _2_bob_   bob_2   BoB
```

There are some reserved words:

```
and, assert, break, class, continue,
def, del, elif, else, except, exec,
finally, for, from, global, if, import,
in, is, lambda, not, or, pass, print,
raise, return, try, while
```

Naming conventions

The Python community has these recommend-ed naming conventions

joined_lower for functions, methods and, attributes

joined_lower or ALL_CAPS for constants

StudlyCaps for classes

camelCase only to conform to pre-existing conventions

Attributes: interface, _internal, __private

Assignment

You can assign to multiple names at the same time

```
>>> x, y = 2, 3
>>> x
2
>>> y
3
```

This makes it easy to swap values

```
>>> x, y = y, x
```

Assignments can be chained

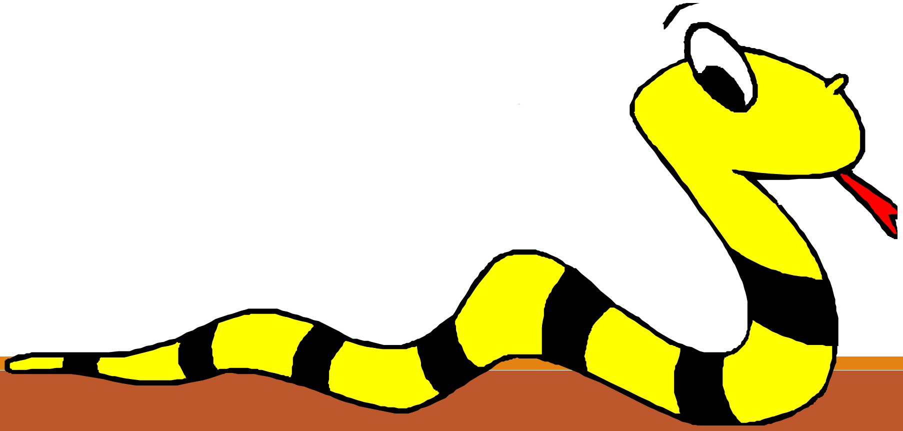
```
>>> a = b = x = 2
```

Accessing Non-Existent Name

Accessing a name before it's been properly created (by placing it on the left side of an assignment), raises an error

```
>>> y

Traceback (most recent call last):
  File "<pyshell#16>", line 1, in -toplevel-
    y
NameError: name 'y' is not defined
>>> y = 3
>>> y
3
```

Sequence types: Tuples, Lists, and Strings



Sequence Types

1. Tuple: (‘john’, 32, [CMSC])
   - A simple *immutable* ordered sequence of items
   - Items can be of mixed types, including collection types
2. Strings: “John Smith”
   - *Immutable*
   - Conceptually very much like a tuple
3. List: [1, 2, ‘john’, (‘up’, ‘down’)]
   - *Mutable* ordered sequence of items of mixed types

Similar Syntax

All three sequence types (tuples, strings, and lists) share much of the same syntax and functionality.

Key difference:

- Tuples and strings are *immutable*
- Lists are *mutable*

The operations shown in this section can be applied to *all* sequence types

- most examples will just show the operation performed on one

Sequence Types 1

Define tuples using parentheses and commas

```
>>> tu = (23, ‘abc’, 4.56, (2,3), ‘def’)
```

Define lists are using square brackets and commas

```
>>> li = [“abc”, 34, 4.34, 23]
```

Define strings using quotes (“, ‘, or “““).

```
>>> st = “Hello World”
>>> st = ‘Hello World’
>>> st = “““This is a multi-line
string that uses triple quotes.”””
```

Sequence Types 2

Access individual members of a tuple, list, or string using square bracket "array" notation

Note that all are 0 based…

```
>>> tu = (23, 'abc', 4.56, (2,3), 'def')
>>> tu[1]     # Second item in the tuple.
 'abc'

>>> li = ["abc", 34, 4.34, 23]
>>> li[1]      # Second item in the list.
 34

>>> st = "Hello World"
>>> st[1]   # Second character in string.
 'e'
```

Positive and negative indices

```
>>> t = (23, 'abc', 4.56, (2,3), 'def')
```

Positive index: count from the left, starting with 0

```
>>> t[1]
'abc'
```

Negative index: count from right, starting with –1

```
>>> t[-3]
4.56
```

Slicing: return copy of a subset

```
>>> t = (23, 'abc', 4.56, (2,3), 'def')
```

Return a copy of the container with a subset of the original members. Start copying at the first index, and stop copying *before* second.

```
>>> t[1:4]
('abc', 4.56, (2,3))
```

Negative indices count from end

```
>>> t[1:-1]
('abc', 4.56, (2,3))
```

Slicing: return copy of a =subset

```
>>> t = (23, 'abc', 4.56, (2,3), 'def')
```

Omit first index to make copy starting from beginning of the container

```
>>> t[:2]
(23, 'abc')
```

Omit second index to make copy starting at first index and going to end

```
>>> t[2:]
(4.56, (2,3), 'def')
```

Copying the Whole Sequence

[:] makes a *copy* of an entire sequence

```
>>> t[:]
(23, 'abc', 4.56, (2,3), 'def')
```

Note the difference between these two lines for mutable sequences

```
>>> l2 = l1 # Both refer to 1 ref,
            # changing one affects both
>>> l2 = l1[:] # Independent copies, two refs
```

The ‘in’ Operator

Boolean test whether a value is inside a container:

```
>>> t = [1, 2, 4, 5]
>>> 3 in t
False
>>> 4 in t
True
>>> 4 not in t
False
```

For strings, tests for substrings

```
>>> a = 'abcde'
>>> 'c' in a
True
>>> 'cd' in a
True
>>> 'ac' in a
False
```

Be careful: the *in* keyword is also used in the syntax of *for loops* and *list comprehensions*

The + Operator

The + operator produces a *new* tuple, list, or string whose value is the concatenation of its arguments.

```
>>> (1, 2, 3) + (4, 5, 6)
 (1, 2, 3, 4, 5, 6)

>>> [1, 2, 3] + [4, 5, 6]
 [1, 2, 3, 4, 5, 6]

>>> "Hello" + " " + "World"
 'Hello World'
```

The * Operator

The * operator produces a *new* tuple, list, or string that "repeats" the original content.

```
>>> (1, 2, 3) * 3
(1, 2, 3, 1, 2, 3, 1, 2, 3)

>>> [1, 2, 3] * 3
[1, 2, 3, 1, 2, 3, 1, 2, 3]

>>> “Hello” * 3
‘HelloHelloHello’
```